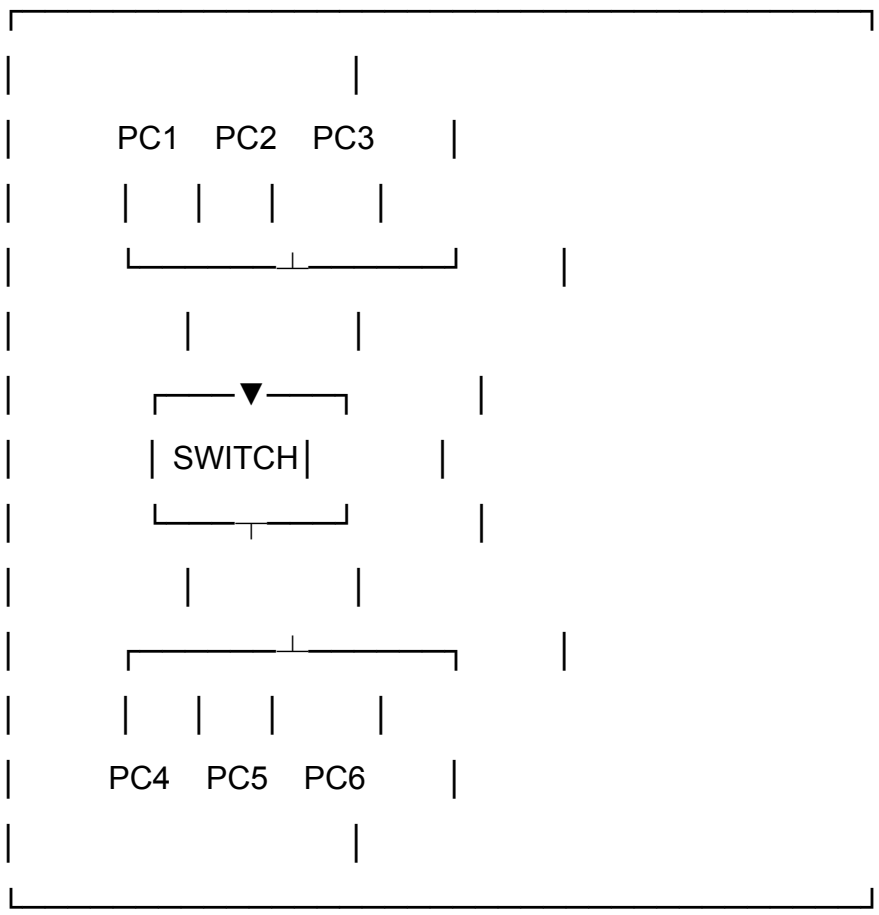
Experiment 3:

Star Topology Design

5.1 Star Topology Overview

In a star topology, all devices are connected to a central device (hub or switch). The central device acts as a communication hub through which all data passes.



5.2 Advantages of Star Topology

- Easy to install and modify
- Centralized management and monitoring
- Fault isolation (failure of one device doesn't affect others)
- Better performance with switches

- Scalable network design

5.3 Disadvantages of Star Topology

- Failure of central device breaks entire network
- Higher cost due to central device
- More cable required

5.4 Step-by-Step Configuration: Star Topology with Switch

1 Open Cisco Packet Tracer and Create New Project

Launch Cisco Packet Tracer

Click File → New

Select "Create Empty Network"

Save the project as "Star_Topology_Switch"

2 Add Network Devices

In the Device Panel, locate the Network Devices section

Click on "Switches" category

Drag a 2960 Switch to the workspace (center position)

Click on "End Devices" category

Drag 6 PCs to the workspace around the switch

Arrange devices in a circular pattern around the switch

3 Connect Devices with Cables

Select the Connection Tool from the toolbar (or press 'C')

Choose "Copper Straight-Through" cable type

Click on PC1 and drag to Switch port Fa0/1

Repeat for PC2 → Fa0/2, PC3 → Fa0/3, etc.

Verify all connections are green (indicating proper connection)

4 Configure IP Addresses for PCs

Click on PC1 to open its configuration window

Go to Desktop tab → IP Configuration

Select "Static" instead of DHCP

Configure as follows:

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.1	255.255.255.0	192.168.1.254
PC2	192.168.1.2	255.255.255.0	192.168.1.254
PC3	192.168.1.3	255.255.255.0	192.168.1.254
PC4	192.168.1.4	255.255.255.0	192.168.1.254
PC5	192.168.1.5	255.255.255.0	192.168.1.254
PC6	192.168.1.6	255.255.255.0	192.168.1.254

5 Test Connectivity

Click on PC1 → Desktop → Command Prompt

Type: ping 192.168.1.2 (to ping PC2)

Verify successful replies

Repeat ping tests between different PCs

Document results in the observation section

Success Indicator: All ping commands should return successful replies with response times typically between 0-1ms in a simulated environment.

5.5 Star Topology with Hub (Alternative Configuration)

1 Create New Project for Hub-based Star

File → New → Create Empty Network

Save as "Star_Topology_Hub"

2 Add Hub and PCs

From Device Panel, click "Hubs" category

Drag a Hub to the center of workspace

Add 6 PCs around the hub

3 Connect and Configure

Connect all PCs to hub ports using straight-through cables

Configure IP addresses (same as switch configuration above)

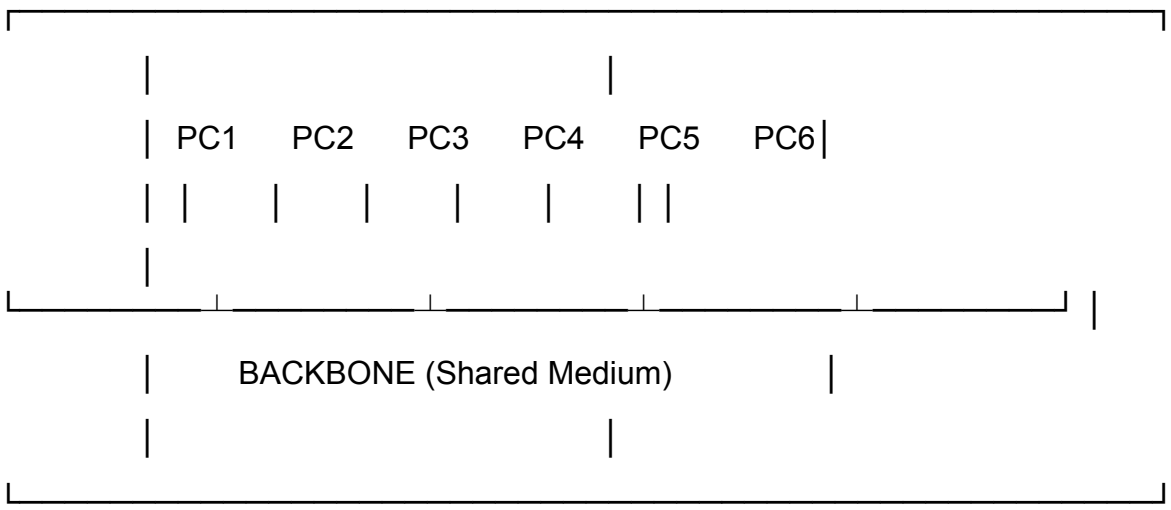
Test connectivity using ping

⚠ **Note:** With a hub, all devices share the same collision domain, resulting in potential collisions and lower performance compared to a switch.

6. Experiment 2: Bus Topology Design

6.1 Bus Topology Overview

In a bus topology, all devices are connected to a single communication line (backbone). Data travels along the bus in both directions until it reaches the destination device.



6.2 Advantages of Bus Topology

- Simple and easy to implement
- Low cost (minimal cabling)
- Easy to extend
- Suitable for small networks

6.3 Disadvantages of Bus Topology

- Limited cable length and number of devices
- Single point of failure (backbone break affects entire network)
- Difficult to troubleshoot

- High collision rate with many devices
- Performance degrades with network traffic

6.4 Step-by-Step Configuration: Bus Topology

1 Create New Project

File → New → Create Empty Network

Save as "Bus_Topology"

2 Add Hub as Backbone

Add 1 Hub to the center of workspace

Add 6 PCs in a linear arrangement

Note: In real bus topology, devices connect to a single cable; in Packet Tracer, we simulate this using a hub

3 Connect Devices in Linear Fashion

Connect all PCs to different ports of the hub

Arrange PCs in a horizontal line to visualize bus topology

Use straight-through cables for connections

4 Configure IP Addresses

Configure all PCs with IP addresses in 192.168.2.0/24 subnet:

Device	IP Address	Subnet Mask
PC1	192.168.2.1	255.255.255.0
PC2	192.168.2.2	255.255.255.0
PC3	192.168.2.3	255.255.255.0
PC4	192.168.2.4	255.255.255.0
PC5	192.168.2.5	255.255.255.0
PC6	192.168.2.6	255.255.255.0

5 Test Connectivity and Observe Collisions

Use ping command to test connectivity between devices

Observe collision indicators on the hub

Send multiple pings simultaneously to observe collision behavior

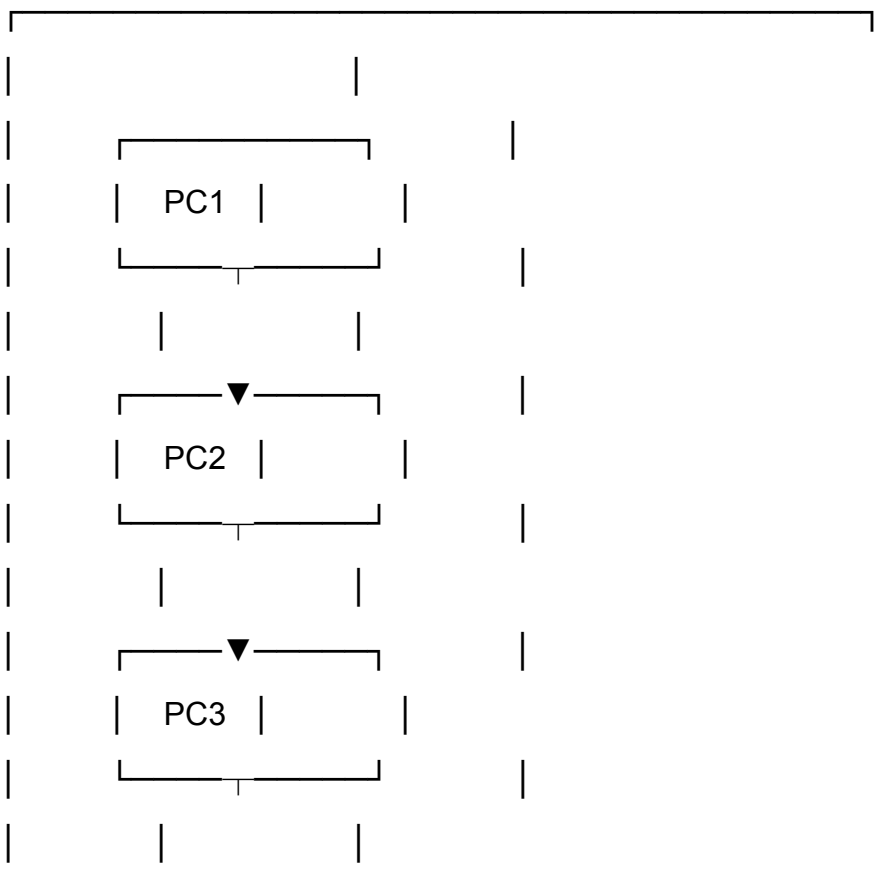
Document observations

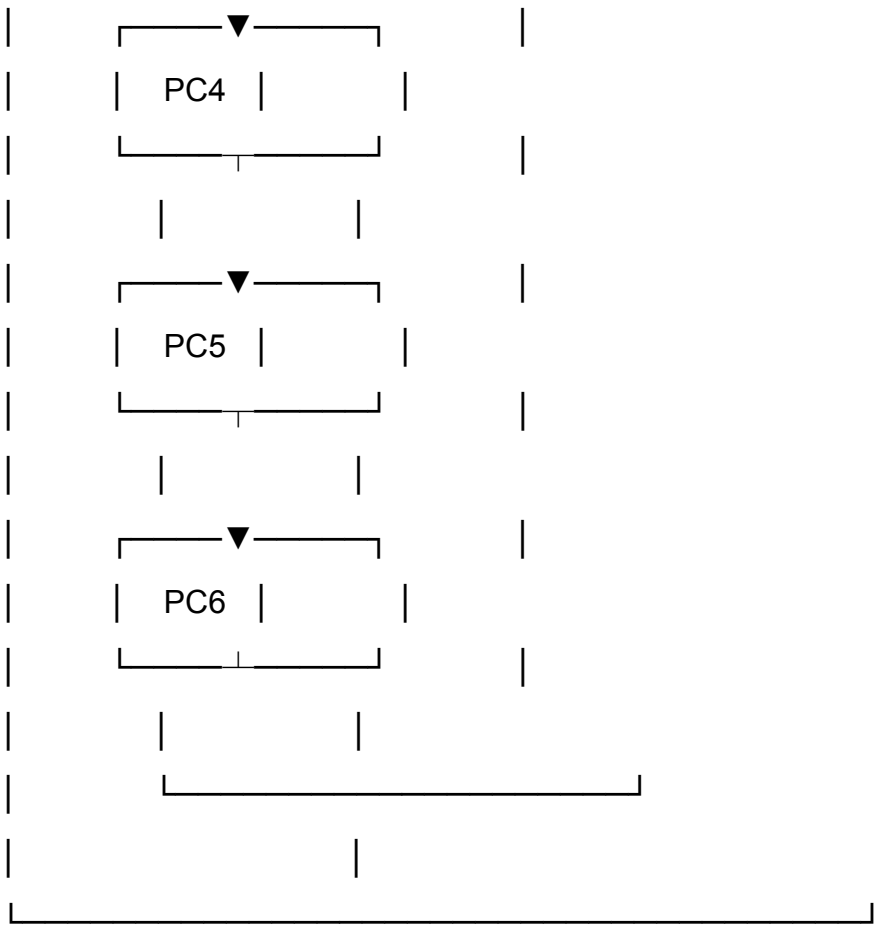
📝 **Note:** In a hub-based bus topology, all devices share a single collision domain. When multiple devices transmit simultaneously, collisions occur, and devices must retransmit their data.

7. Experiment 3: Ring Topology Design

7.1 Ring Topology Overview

In a ring topology, each device is connected to exactly two other devices, forming a closed loop. Data travels in one direction (or both directions in bidirectional ring) around the ring.





7.2 Advantages of Ring Topology

- Fair access to network resources
- Predictable performance
- No collision domain issues
- Suitable for token-based protocols

7.3 Disadvantages of Ring Topology

- Break in ring breaks entire network
- Difficult to add or remove devices
- More complex than star topology
- Higher latency due to sequential data passing

7.4 Step-by-Step Configuration: Ring Topology

1 Create New Project

File → New → Create Empty Network

Save as "Ring_Topology"

2 Add Switches and PCs

Add 6 Switches (2960) to the workspace in a circular arrangement

Add 6 PCs, each connected to one switch

Arrange devices in a circular pattern

3 Connect Switches in Ring Formation

Connect Switch1 port Fa0/2 to Switch2 port Fa0/1

Connect Switch2 port Fa0/2 to Switch3 port Fa0/1

Connect Switch3 port Fa0/2 to Switch4 port Fa0/1

Connect Switch4 port Fa0/2 to Switch5 port Fa0/1

Connect Switch5 port Fa0/2 to Switch6 port Fa0/1

Connect Switch6 port Fa0/2 to Switch1 port Fa0/1 (completing the ring)

4 Connect PCs to Switches

Connect PC1 to Switch1 port Fa0/3

Connect PC2 to Switch2 port Fa0/3

Connect PC3 to Switch3 port Fa0/3

Connect PC4 to Switch4 port Fa0/3

Connect PC5 to Switch5 port Fa0/3

Connect PC6 to Switch6 port Fa0/3

5 Configure IP Addresses

Configure all PCs with IP addresses in 192.168.3.0/24 subnet:

Device	**IP Address**	**Subnet Mask**
PC1	192.168.3.1	255.255.255.0
PC2	192.168.3.2	255.255.255.0
PC3	192.168.3.3	255.255.255.0

PC4	192.168.3.4	255.255.255.0
PC5	192.168.3.5	255.255.255.0
PC6	192.168.3.6	255.255.255.0

6 Test Connectivity

Ping between adjacent devices (PC1 to PC2, PC2 to PC3, etc.)

Ping between non-adjacent devices (PC1 to PC4, PC1 to PC6)

Observe how data travels through the ring

Document response times and path taken

7 Test Ring Failure Scenario

Disconnect one link in the ring (e.g., between Switch3 and Switch4)

Attempt to ping PC3 from PC4

Observe that connectivity is lost

Reconnect the link and verify connectivity is restored